Data Science in Healthcare- Resources and Timeline

- ## Week 1: Exploratory Data Analysis-

  Replace NaN values with column means, convert time series data from 1 value per millisecond to 1 value per second using “numpy decimate”, concatenate 8 hour’s data for all 4 patients. Do basic EDA: plot some channel variations, correlation metrics, normalize the data etc. please note to use “mne” library to load . edf files. (This is a bit open ended on the EDA part; you can find your own ways to visualize data). Make a habit to save the filtered data whenever you have it since we are dealing with big data, your sessions can crash in between the computation. Data- WiDS Data (ignore hour 9 in some folders).

- ## Week 2: Identifying Relevant channels-

  For every patient, for every channel, find the channel mean across the 8 hour data and to see how this value is varying across different patients, find mean (a) and standard deviation (b) of the 4 means for 4 patients respectively. The value we are using as varying factor here is (a/b). Do further EDA on the mean data that you have now. The format for this data should be: (34 x 7 matrix)

channel	Normal mean	Mild mean	Moderate mean	Severe mean	Mean of means (a)	Std dev of means (b)	Varying factor (a/b)
Sp02							
EEG							
ECG and so on …							

  The further EDA can focus on plotting channel variations across different patients, doing some PCA to show how some channels are correlated with each other (if time permits). Finally using the varying factor, get top 8-10 channels which we will consider as the relevant channels in our further analysis. (note that channels named Flow Patient- 0, Flow Patient- 1, Sp02 should be there for further analysis despite the fact that they were among top “relevant channels” or not).

- ## Week 3: Identifying Critical points in the filtered time series data-

  Now to actually classify a patient as one of the 4 classes, there is a study done on how to find some critical points- “Apnea Points” and “Hypopnea Points”. Go through the study given below to understand more about these points- [Sleep Education for Sleep Ed Pulmonary Fellows Modules](). If the above is overwhelming, go through the ppt- OSA_Notes.pdf . To know more about what our data channels mean, have a look at this sheet- Channel definitions . After the literature review, code methods to find these critical points across the data (use time series data with 1 second per value). Observe how the AHI index is varying across patients. Now, plot the variation in the all “relevant channels” 100 seconds before and after these points. Also compute percent variation in these channels (between mean from x-100 to x and mean from x to x+100 where x is the critical point. Do this for both apnea and hypopnea points). Now average this percent variation across all critical points and arrange the data to know which channels are getting affected the most near the critical points. (Open for further analysis if time permits)

- **Week 4: Machine Learning Implementation (optional)-**
  Mostly this will be buffer week but for those who have completed previous stuff on time, this week we will focus on ML implementation. Observing the data closely, you would realise that patient is not awake all the time in some cases, and hence it changes a lot of dynamics. Our further aim is to leverage methods used previously to the data only at specific times. Will have a detailed meet to describe what exactly can be done this week.

Tentative Timeline-

- Week 1: wrap up by 23th December
- Week 2: 24th Dec - 31st Dec
- Week 3: 1st Jan- 6th Jan
- Week 4: 7th Jan- 13th Jan

Final deliverables- Code (.ipynb preferred) and a brief report about the methodologies used.

For any further issues, feel free to put it up in the whatsApp chat or DM the mentors-

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In case, Analytics Club asks for midterm review, you will be notified about the midterm deliverables soon. Also the data provided is copyrighted and hence use it only for this project.